Neural-Surrogate-Guided Asteroid Lightcurve Inversion: Shape Reconstruction and Photometric Degeneracy

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Introduction

Lightcurve Inversion

Can we reconstruct asteroid shapes from disk-integrated lightcurves?

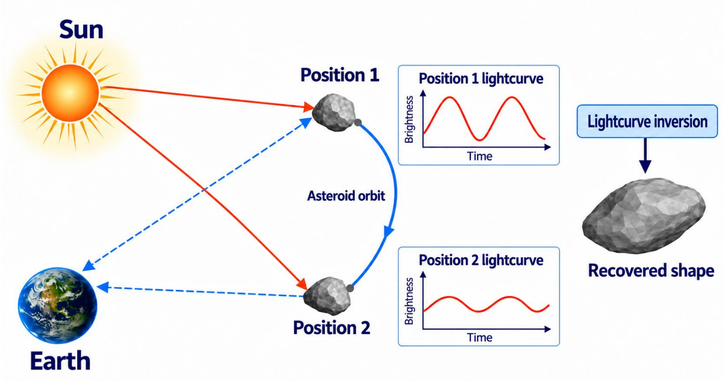


Fig. 1. Different Sun–asteroid–Earth observing geometries produce distinct rotational lightcurves(LC). Combining lightcurves obtained at multiple orbital positions enables the reconstruction of the asteroid's three-dimensional shape.

Sequential Reformulation

- on $\theta - \phi$ grid polyhedron : discrete action space

$$\boldsymbol{r}_{(0)} \to \boldsymbol{r}_{(1)} \to \cdots \to \boldsymbol{r}_{*} = \mathrm{argmin}_{\boldsymbol{r}} \mathcal{J}(\boldsymbol{r})$$

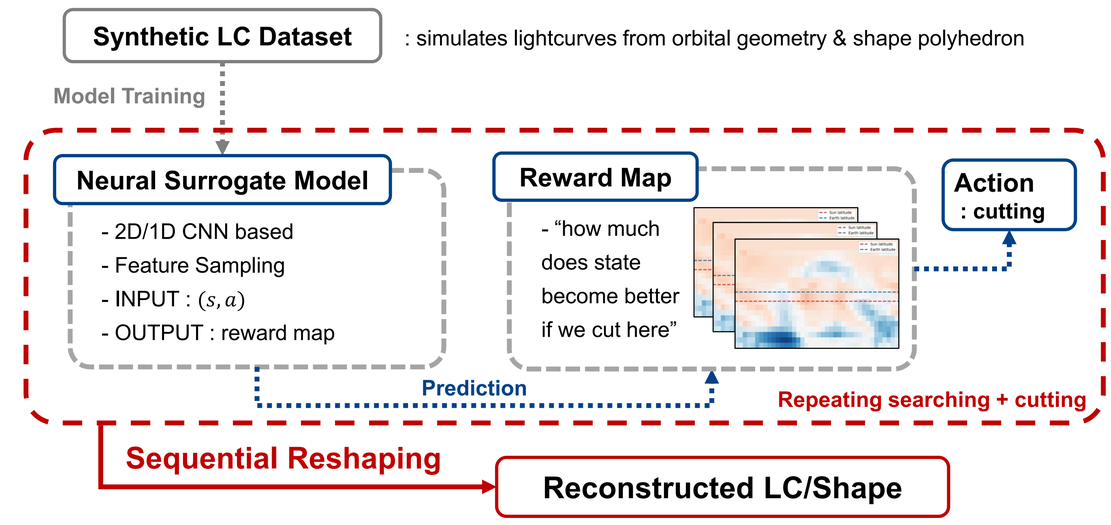
- Instead of optimizing the full shape at once, we reconstruct it through a sequence of local deformations.

Neural-Surrogate-Guided Reconstruction

Pipeline

- state $s = \mathrm{r}_{\mathrm{arr}} + \mathrm{LC}_{\mathrm{target}} + \mathrm{LC}_{\mathrm{model}} + \mathrm{LC}_{\mathrm{info}}$
- action $a = (\phi_a, \theta_a)$

Fig. 2. Overview of the neural-surrogate-guided sequential reshaping pipeline. A trained surrogate predicts reward maps to guide iterative shape reconstruction.



Performance

- Most samples achieve low lightcurve error, indicating reliable reconstruction in the majority of cases.
- The remaining high-error tail reflects difficult or strongly degenerate cases.

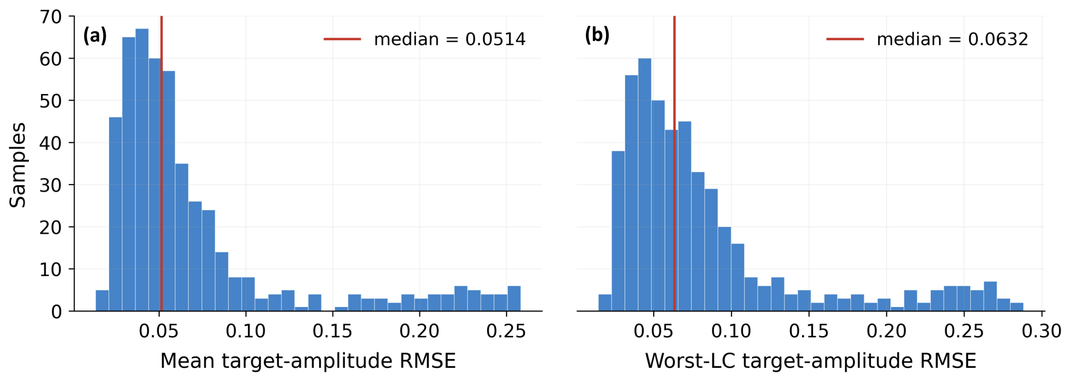


Fig. 3. Reconstruction-error distributions for the best recovered states: (a) sample-mean target-amplitude RMSE and (b) worst-lightcurve target-amplitude RMSE. Red vertical lines indicate the median values, 0.0514 and 0.0632, respectively.

Representative Reconstruction

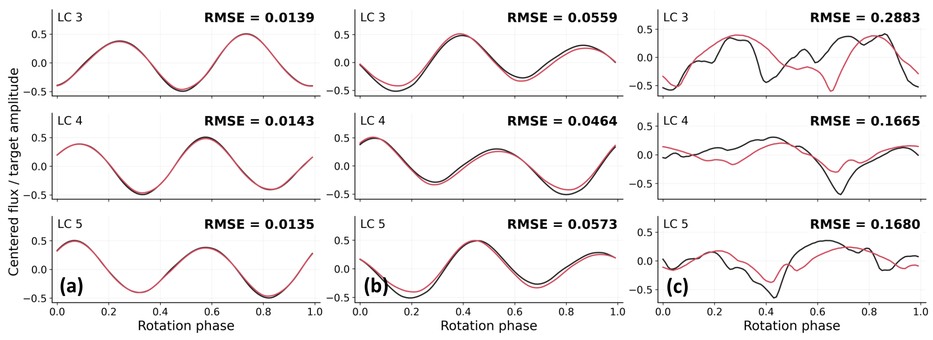


Fig. 4. Lightcurve reconstruction examples at (a) low, (b) moderate, and (c) high fitting errors. Black and red curves represent target and reconstructed lightcurves, respectively.

Geometry Dependence

But a good LC fit does not imply a unique shape.
→ What shape variations remain unconstrained?

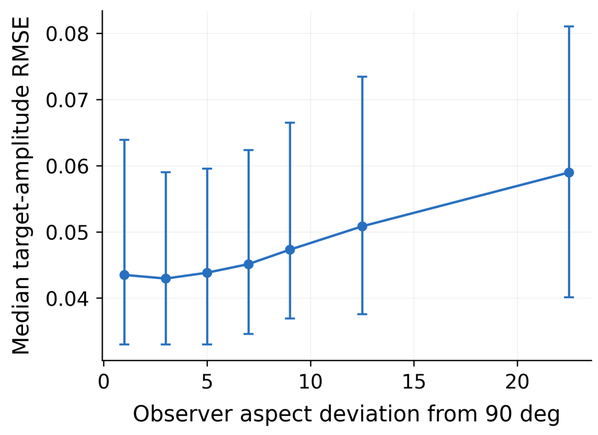


Table: Metric Scale

NRMSE range	Interpretation
0.02 – 0.05	Excellent
0.05 – 0.10	Good
0.10 – 0.20	Moderate
0.20 – 0.40	Mismatch
> 0.40	Poor

- Reconstruction is most accurate for near-equator viewing geometries and degrades away from 90° observer aspect.

Fig. 5. Median training lightcurve RMSE versus observer-aspect deviation from 90°. Reconstruction error generally increases with larger deviations.

Photometric Degeneracy

Mathematical Formulation

- forward map $\mathcal{F}: \mathcal{R} \to \mathcal{L}$, $\mathbf{r} \mapsto \mathbf{L}(\mathbf{r}; \Gamma)$
- photometrically equivalent family $\mathcal{M}(\mathbf{r}_0) = \{\mathbf{r} \mid \mathcal{F}(\mathbf{r}) = \mathcal{F}(\mathbf{r}_0)\} \subseteq \mathcal{R}$
- photometrically feasible family $\mathcal{S}_\epsilon(\mathbf{r}_0) = \{\mathbf{r} \mid d_{\mathrm{LC}}(\mathcal{F}(\mathbf{r}), \mathcal{F}(\mathbf{r}_0)) \le \epsilon\} \supset \mathcal{M}(\mathbf{r}_0)$
  : *we seek shape deformations that leave the lightcurve nearly unchanged.*
- if locally $J_{\mathcal{F}}(\boldsymbol{r}_0)\delta\mathbf{r} \approx 0$, $\delta\mathbf{r}$ 1st –order photometric null direction

Feasible Shape Family Construction

Sequential inversion generates an ensemble of photometrically feasible shapes.

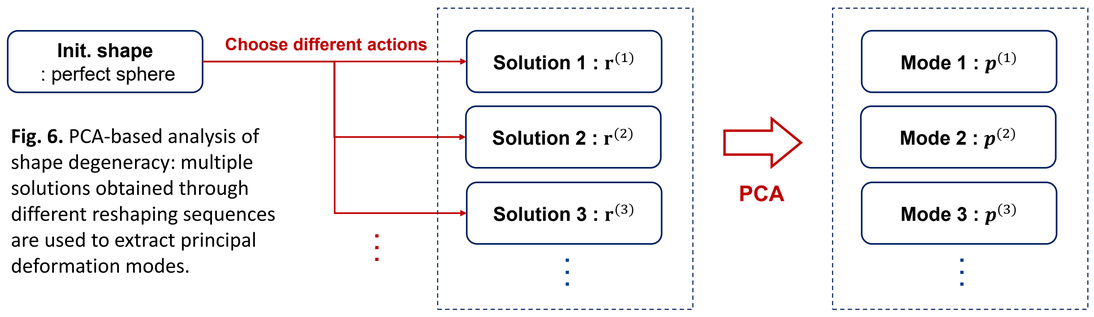


Fig. 6. PCA-based analysis of shape degeneracy: multiple solutions obtained through different reshaping sequences are used to extract principal deformation modes.

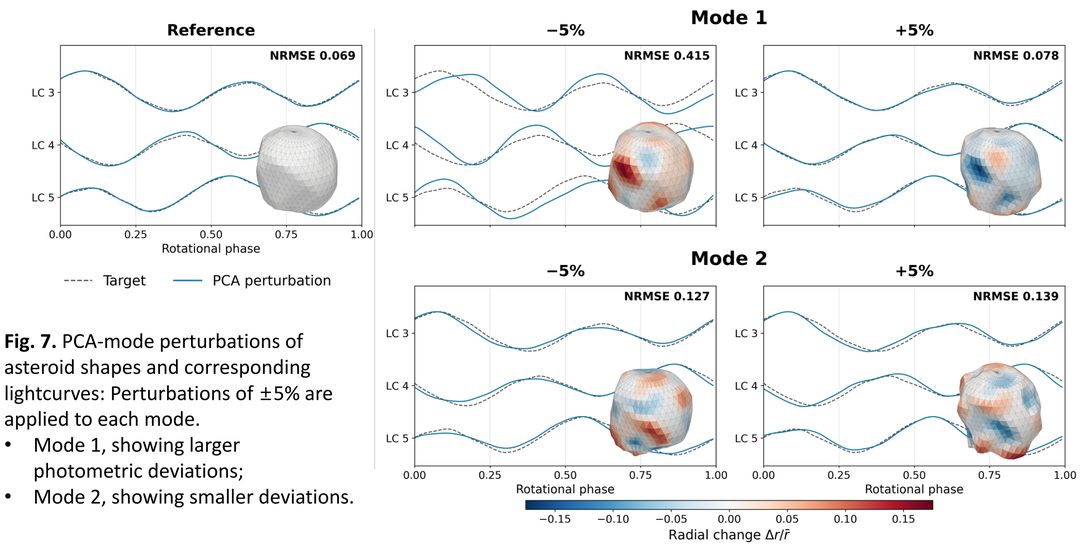


Fig. 7. PCA-mode perturbations of asteroid shapes and corresponding lightcurves: Perturbations of ±5% are applied to each mode.
- Mode 1, showing larger photometric deviations;
- Mode 2, showing smaller deviations.

- Different PCA modes produce substantial shape variations with only small photometric changes.
- Collecting PCA modes, we constructed empirical feasible deformation subspace:

$$V_{\mathrm{PCA}} = \mathrm{span}\{\mathbf{p}_1, \cdots, \mathbf{p}_k\}$$

Higher-Order Sensitivity

Q. What did we find from PCA?

$\mathcal{S}_\epsilon(\mathbf{r}_0) \approx \mathbf{r}_0 + V_{\mathrm{PCA}}$? : NO

$$\mathcal{F}(\mathbf{r}_0 + c\mathbf{p}) \approx \mathcal{F}(\mathbf{r}_0) + cJ_{\mathcal{F}}(\mathbf{r}_0)\mathbf{p} + \frac{c^2}{2} D^2\mathcal{F}(\boldsymbol{r}_0)[\mathbf{p}, \mathbf{p}]$$

- Photometric null modes are defined in the infinitesimal regime.
- $\mathbf{p}$ gives finite-amplitude perturbation.

PCA-derived deformation **modes remain photometrically flatter** than first-order-matched random directions even at finite amplitude.

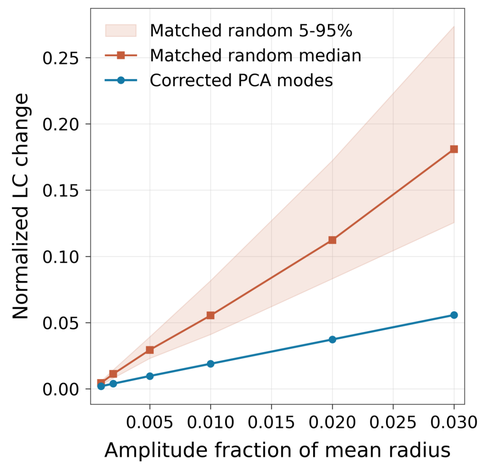


Fig. 8. Normalized lightcurve changes under PCA-mode and matched random perturbations. Corrected PCA modes exhibit substantially smaller photometric variations at the same perturbation amplitude.

Conclusion

Sequential reshaping turns lightcurve inversion from finding a shape into probing *which shape degrees of freedom are actually constrained by photometry.*

Summary

1. Efficient inversion
Neural surrogate guidance enables sequential asteroid-shape reconstruction from lightcurves.

2. From a solution to a solution family
Different reshaping trajectories expose multiple photometrically feasible shapes.

3. Probing the information content of lightcurves
PCA reveals empirical weak deformation modes that remain unusually insensitive even beyond the infinitesimal regime.

Caveats

- Results are currently based primarily on synthetic data.
- Degeneracy depends on observing geometry and model assumptions.

Future Work

- Validate the framework on real asteroid lightcurves and broader observing geometries.
- Quantify how additional geometries reduce the feasible deformation space.

References
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[2] Kaasalainen, M., Torppa, J., & Muinonen, K. 2001, *Icarus*, **153**, 37–51.
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